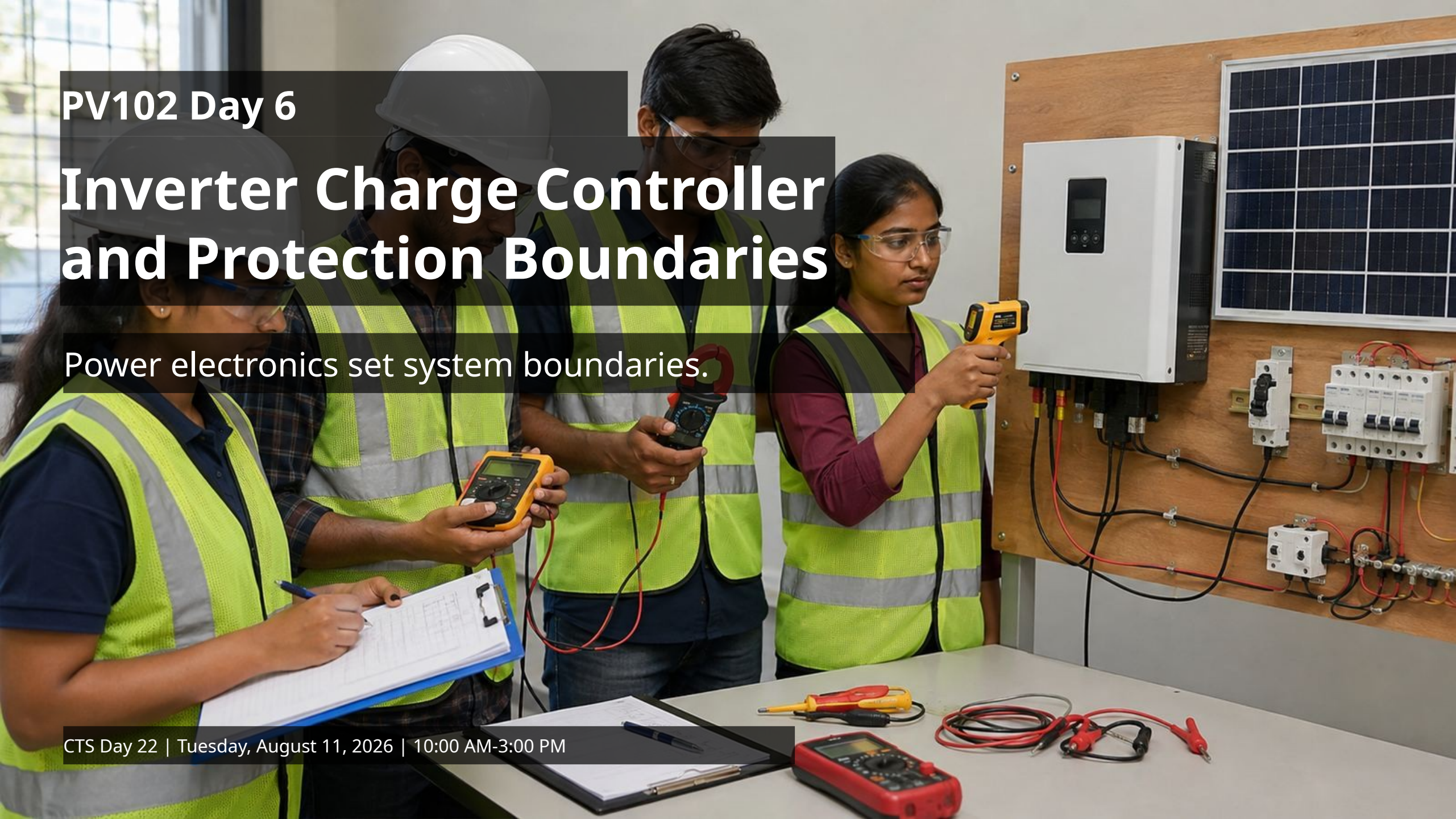


PV102 Day 6

Inverter Charge Controller and Protection Boundaries

Power electronics set system boundaries.

CTS Day 22 | Tuesday, August 11, 2026 | 10:00 AM-3:00 PM

Today’s job

Professional solar work starts before parts move.

• Students map the balance-of-system decisions that make PV energy usable and controlled: charge controllers, inverters, disconnects, overcurrent protection, grounding/bonding boundaries, and labeled handoff points.

- Carry forward the Day 1 habit: match architecture to real need.
- Today adds one design layer and one clean handoff.

Do not guess when an unknown can be recorded.



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Learning targets

By 3:00 PM students should be able to:

- Explain the roles of charge controllers and inverters in battery-based PV systems.
- Match inverter selection to continuous load, surge load, voltage, and output requirement.
- Identify where disconnects and overcurrent protection belong conceptually.
- Create a block diagram that separates source, storage, conversion, and load boundaries.



Evidence beats confidence. A tragic lesson for civilization, apparently.

Power electronics set system boundaries.

Morning readiness check

Quiz first, then review as evidence.

- Work independently for 12-15 minutes.
- Use wrong answers to locate teaching gaps.
- Ask: what fact would prove this answer?
- Keep review short and specific.



The quiz measures instruction as much as students.

Core rule

- Power electronics set system boundaries.
- Say it once at the start.
- Bring every debate back to it.
- Use it to reject weak handoffs.



One rule, repeated until reality behaves.

What students already know

PV101 and RCS are not separate islands.

- PV modules produce DC power.
- Series and parallel change voltage/current behavior.
- Continuity requires de-energized verification.
- Troubleshooting starts from evidence.
- Handoffs protect the next person.



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New judgment today

The skill moves from “can I build it?” to “should this be the design?”

- Define the job.
- Gather the evidence.
- Name the constraint.
- Protect the boundary.
- Record unknowns.



Power electronics set system boundaries.

Energy path and decision boundary



Every arrow is a boundary. Boundaries need ratings, protection, isolation, and documentation.

Decision checks

Decision	Evidence	Reject if
Keep a design choice	Measured or observed site/load data	It only sounds good
Reject a shortcut	Safety or equipment boundary	It hides a hazard
Carry unknown forward	Named verification step	It becomes fake certainty
Approve handoff	Next person can act safely	It depends on mind reading

Guided example

Model one complete thinking path.

- Start with the client/job statement.
- Add one measured or observed fact.
- Make one design decision.
- Name one safety boundary.
- Name one unknown and how to verify it.



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Team roles

Three teams of two, roles rotate after the first pass.

- Designer: proposes the first path.
- Verifier: challenges evidence and limits.
- Recorder/handoff lead: writes what the next person needs.
- In a pair, one person may hold two roles, then swap.



Lab mission

Balance-of-System Block Diagram

- Teams build a labeled block diagram and defend every protection/isolation point.



This is the work block. Keep it practical and evidence-based.

Scenario options

Assign one, or let teams choose if the room can handle democracy today.

- 12 V training system with PV module, controller, LiFePO4 battery, inverter, DC load, AC load.
- 48 V bus system concept with hybrid inverter/charger, battery bank, PV strings, AC distribution.
- Remote DC-only sensor system with no inverter and limited service visits.



Power electronics set system boundaries.

Lab procedure

Run the same loop every day.

- Choose a scenario and place component cards in energy-path order.
- Draw source, storage, conversion, and load boundaries.
- Add conceptual OCPD/disconnect labels where energy could feed a fault or where service isolation is needed.
- Mark manufacturer-instruction-required decisions.
- Run a peer safety check: can someone tell what is energized, isolated, and unknown?



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Acceptance criteria

A complete answer has five parts.

- Match the need.
- Show the path.
- Name the dominant constraint.
- State the safety boundary.
- Record unknowns instead of guessing.



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Safety boundary

No training objective is improved by pretending energy is harmless.

- Use only instructor-approved equipment and procedures.
- Treat PV and batteries as energy sources.
- Stop when measurements do not make sense.
- Follow manufacturer instructions and program safety procedures.

Stop, isolate, verify, ask. The glamorous life.



Power electronics set system boundaries.

Instructor probe questions

Use these when teams get vague.

- What evidence supports that?
- What would make this design fail?
- Which boundary protects the person or equipment?
- What unknown are you carrying forward?
- What would the next crew need to know?



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Peer verification

Peers check the handoff, not the personality of the speaker. Humanity struggles here.

- Can I identify the job?
- Can I follow the path?
- Can I name the constraint?
- Can I see the safety boundary?
- Can I find the unknowns?



Power electronics set system boundaries.

Common failure modes

Correct these early.

- Component list with no job.
- Guess presented as fact.
- Missing safety boundary.
- No units or runtime.
- Handoff too vague to use.



Evidence language

Require technical sentences.

- Observed: what you saw.
- Measured: value with unit or clear method.
- Assumed: guess being carried forward.
- Verified: checked against source, procedure, or measurement.



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Handoff structure

Make the next person safer.

- Scope: what job is being solved.
- Decision: what was chosen or rejected.
- Evidence: why.
- Boundary: what must be protected.
- Unknowns: what still needs verification.



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Exit ticket

Individual close-out

- Which boundary on your diagram mattered most for safety, and what would happen if it were missing?



Collect before reset so the day leaves a paper trail. Weirdly useful.

Day-to-day continuity

This is one system, not seven disconnected worksheets.

- PV102 Day 7 turns the design into an install plan, tool/material list, JHA, and daily work sequence.
- Carry forward site notes, load tables, diagrams, and unknowns.
- Do not erase uncertainty. Resolve it.



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Final reminder

What professionals do

- Define the job.
- Verify the path.
- Protect the boundary.
- Document the truth.



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Close

- Power electronics set system boundaries.



Reset tools. Collect packets. Archive the handoffs.